

DAY 3

Easy :

```
class Solution:
    def mergeTwoLists(self, list1: Optional[ListNode], list2:
Optional[ListNode]) -> Optional[ListNode]:
        dummy = ListNode()
        current = dummy

        while list1 and list2:
            if list1.val < list2.val:
                current.next = list1
                list1 = list1.next
            else:
                current.next = list2
                list2 = list2.next
            current = current.next

        if list1:
            current.next = list1
        elif list2:
            current.next = list2

        return dummy.next
```

☒ Testcase | ☒ Test Result

Accepted Runtime: 0 ms

☒ Case 1 ☒ Case 2 ☒ Case 3

Input

list1 =
[1,2,4]

list2 =
[1,3,4]

Output

[1,1,2,3,4,4]

Expected

[1,1,2,3,4,4]

Medium :

```
class Solution:
    def productExceptSelf(self, nums: List[int]) -> List[int]:
        n = len(nums)
        answer = [1] * n

        left_product = 1
        for i in range(n):
            answer[i] = left_product
            left_product *= nums[i]

        right_product = 1
        for i in range(n - 1, -1, -1):
            answer[i] *= right_product
            right_product *= nums[i]

        return answer
```

Testcase | Test Result

Accepted Runtime: 0 ms

Case 1

Case 2

Input

nums =
[1,2,3,4]

Output

[24,12,8,6]

Expected

[24,12,8,6]